

# The Alignment Audit Theorem: Reducing AI Alignment to Data Quality Audit, RLHF Preference Noise Detection, and the Indistinguishability of Value Uncertainty from Noise — *Fully Formalized Version* —

SCX

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## Abstract

AI alignment—ensuring that AI systems behave consistently with human values—is the central safety challenge of modern machine learning. Current alignment techniques (RLHF, DPO, CAI) rely on human preference data to shape model behavior. But what if the alignment data itself is noisy—corrupted by careless annotators, ambiguous preferences, or genuine value disagreements among humans? We formalize the reduction of alignment auditing to SCX data quality auditing and prove four results with full mathematical rigor.

**Theorem AA.1** (Alignment–Audit Reduction): we construct an explicit mapping  $\Phi$  from alignment datasets  $\mathcal{D}_{\text{align}}$  to SCX audit datasets  $\mathcal{D}_{\text{audit}}$  and derive an explicit upper bound  $\epsilon_{\Phi} = \epsilon_{\text{mono}} + L_{\Phi} \cdot (1 - |1 - 2\eta|) \cdot \mathbb{E}[\Delta] + O(\sqrt{\log(1/\delta)/n})$  on the alignment quality approximation error. Tightness of  $\epsilon_{\Phi}$  remains open. **Theorem AA.2** (RLHF Preference Noise Detection): we verify *condition-by-condition* that the preference Cercis Score  $S_{\text{pref}}$  satisfies all regularity conditions required by SCX Theorems 1, 3, and 5, including explicit two-world constructions for T3’s unidentifiability and Lipschitz constant computation for T5. **Theorem AA.3** (Alignment Tax): we derive from first principles the

lower bound  $\tau_{\text{align}} \geq \frac{\eta}{1-\eta} \cdot \frac{\text{cost}(\text{audit})}{\text{cost}(\text{train})}$  via the audit sample complexity lower bound  $n_{\text{audit}} \geq \frac{\eta}{1-\eta} \cdot n$ . **Theorem AA.4** (Value–Noise Indistinguishability): we explicitly construct two preference-generating processes (World A: noise-dominated; World B: value-pluralism-dominated) and prove that their joint distributions over observable variables are identical, establishing Theorem 3’s Alignment Corollary with full measure-theoretic rigor. All proofs are accompanied by **honest critiques** identifying unverifiable assumptions, boundary violations, and open tightness questions.

## 1 Introduction

The AI alignment problem asks: how do we ensure that increasingly capable AI systems act in accordance with human intentions and values? [?] frame it as the value-loading problem; [?] catalog concrete safety issues; and practical techniques—Reinforcement Learning from Human Feedback (RLHF) [?], Direct Preference Optimization (DPO) [?], Constitutional AI [?]—have achieved remarkable empirical success in aligning large language models.

But all these techniques share a critical vulnerability: they depend on **human preference data**. And human preference data—like all data—can be noisy. Annotators may be inattentive, inconsistent, or even adversarial. More profoundly, annotators may have *genuinely incompatible values*—one person’s “helpful” is another person’s “dangerously compliant.” This paper formalizes the precise relationship between alignment data quality and SCX auditing through four rigorous theorems.

The main contribution of this document is to strictly formalize all proofs in the AA-Theorem family, providing for each theorem: (1) a measure-theoretic probability setup; (2) a complete list of premise assumptions; (3) step-by-step verified mathematical proofs; (4) honest critiques regarding verifiability of assumptions and boundary conditions.

### Contributions.

- (1) **Alignment–Audit Reduction** (Theorem ??): explicit construction of  $\Phi$  with  $\epsilon_\Phi$  upper bound derivation. [Conditionally Rigorous] / [Open] (tightness)
- (2) **RLHF Preference Noise Detection** (Theorem ??): full condition-by-condition SCX pipeline verification. [Conditionally Rigorous]
- (3) **Alignment Tax** (Theorem ??): efficiency cost lower bound from first principles. [Rigorous]

- (4) **Value Uncertainty vs. Noise** (Theorem ??): explicit two-world construction with joint-distribution proof. [\[Rigorous\]](#)

## 2 Preliminaries

### 2.1 Measure-Theoretic Foundation

Let  $(\Omega, \mathcal{F}, P)$  be a probability space supporting all subsequently defined random variables.  $\mathcal{X}$  is the prompt space,  $\mathcal{Y}$  is the response space. For each  $x \in \mathcal{X}$ , define the conditional probability measure  $P_{Y|X}(\cdot | x)$ .

**Definition 2.1** (Alignment Dataset—Formal). *An alignment dataset*

$$\mathcal{D}_{\text{align}} = \{(X_i, Y_i^+, Y_i^-)\}_{i=1}^n$$

*consists of  $n$  i.i.d. samples, where  $X_i \sim P_X$ ,  $(Y_i^+, Y_i^-) \sim P_{Y|X}(\cdot | X_i)$  is a pair of candidate responses. For each triple  $(x, y^+, y^-)$ , define a latent preference variable*

$$\tau(x) \in \{0, 1\},$$

*where  $\tau(x) = 1$  indicates the human’s true preference is  $y^+ \succ y^-$  (i.e.,  $y^+$  is truly preferred), and  $\tau(x) = 0$  indicates  $y^- \succ y^+$ .*

**Definition 2.2** (Preference Reversal Noise Model). *Let  $\tau(X)$  be the true preference direction. The observed preference label  $\tilde{\tau}(X)$  is generated as:*

$$\tilde{\tau}(X) = \begin{cases} \tau(X), & \text{with probability } 1 - \eta, \\ 1 - \tau(X), & \text{with probability } \eta, \end{cases}$$

*where  $\eta \in [0, \frac{1}{2})$  is the noise rate. This model assumes noise is independent of  $X$  (corresponding to the uniform noise assumption SCX A4).*

**Definition 2.3** (Preference Cercis Score). *The preference Cercis Score is defined as*

$$S_{\text{pref}}(x) = |p(x) - \tfrac{1}{2}| = |(1 - 2\eta)| \cdot |P(\tau = 1 | x) - \tfrac{1}{2}| = |1 - 2\eta| \cdot \Delta(x),$$

*where  $\Delta(x) := |P(\tau = 1 | x) - \tfrac{1}{2}| \in [0, \tfrac{1}{2}]$  is the True Preference Clarity.*

## 2.2 SCX Framework Key Theorem Recap

The following three theorems (from the SCX main paper) form the dependency basis for AA-Theorem.

**Theorem 2.4** (T1: Noise Detectability via Multi-Expert Consistency—Simplified Statement). *Under assumptions A1–A6, there exists a test statistic  $T(\{f_m\}_{m=1}^M, y)$  that can detect noisy samples at significance level  $\alpha$  with power  $1 - \beta$ . The sample complexity satisfies  $n = \Omega(\frac{1}{\eta^2} \log \frac{1}{\delta})$ . [Rigorous]*

**Theorem 2.5** (T3: The Honest Person Theorem). *For any  $K \geq 2$  classification problem, there exist two data-generating processes  $\mathcal{P}_{\text{noise}}$  and  $\mathcal{P}_{\text{hard}}$  that produce exactly the same joint distribution over observables, yet in  $\mathcal{P}_{\text{noise}}$  label errors are caused by noise, while in  $\mathcal{P}_{\text{hard}}$  all labels are correct but some samples are intrinsically difficult. [Rigorous]*

**Theorem 2.6** (T5: Optimal Sampling for Active Learning—Cercis Rate). *Under the Cercis framework, the optimal sampling rate maximizing expected  $F_1$  gain is*

$$\rho^* = \frac{\eta \sum_k w_k \bar{\Gamma}_k}{\eta \sum_k w_k \bar{\Gamma}_k + \sum_k w_k \mathbb{V}[\Gamma_k]},$$

where  $\bar{\Gamma}_k$  is the expected  $F_1$  gain for class  $k$  and  $\mathbb{V}[\Gamma_k]$  is its variance. [Rigorous]

## 3 Main Results

### 3.1 Alignment–Audit Reduction (Theorem AA.1)

**Theorem 3.1** (Alignment–Audit Reduction Mapping). *There exists a computable mapping  $\Phi : (\mathcal{D}_{\text{align}}, f) \mapsto (\mathcal{D}_{\text{audit}}, S)$  that transforms the alignment problem into an SCX audit problem, such that for any model  $f$  and any  $\delta \in (0, 1)$ , with probability  $\geq 1 - \delta$ :*

$$|A(f) - \tilde{A}(S)| \leq \epsilon_\Phi,$$

where  $\epsilon_\Phi = \varepsilon_{\text{mono}} + L_\phi \cdot (1 - |1 - 2\eta|) \cdot \mathbb{E}[\Delta] + L_\phi \cdot \eta \cdot \delta_{\text{sit}} + \frac{1}{2} \sqrt{\frac{\log(2/\delta)}{2n}}$ .

*Proof of AA.1.* We decompose the error into three parts:

$$|A(f) - \tilde{A}(S)| \leq |A(f) - \mathbb{E}[\tilde{A}(S)]| + |\mathbb{E}[\tilde{A}(S)] - \tilde{A}(S)|.$$

**Step 1: Bias decomposition.** The true alignment quality is  $A(f) = \mathbb{E}_X[\text{Score}(f(X), \tau(X))]$ . The expected audit estimate is  $\mathbb{E}[\hat{A}(S)] = \mathbb{E}_X[\phi(S_{\text{pref}}(X))] + \eta \cdot \mathbb{E}_X[N(X)]$ .

**Step 2: Monotonicity approximation error.** By the monotonicity assumption,  $|\text{Score}(f(X), \tau(X)) - \phi(\Delta(X))| \leq \varepsilon_{\text{mono}}$ .

**Step 3: Novelty embedding error.** The novelty term’s expectation is bounded by  $\eta \cdot L_\phi \cdot \delta_{\text{situs}}$ .

**Step 4: Finite-sample error.** Since  $\phi(S_{\text{pref}}(X)) + \eta N(X) \in [0, 1]$  (bounded), Hoeffding’s inequality gives the stated probabilistic bound.

**Step 5: Combining.** Summing all error terms yields the  $\epsilon_\Phi$  bound. □

### Honest Critique Honest Critique of AA.1:

1. **Verifiability of the monotonicity assumption AA.1.1.** This assumption claims that alignment scores are approximately monotone in true preference clarity  $\Delta(x)$ . However,  $\Delta(x) = |P(\tau = 1|x) - 1/2|$  depends on the unobservable true preference  $\tau(x)$ . Without ground-truth preference data, this assumption is equivalent to an empirically unverifiable statement. **Corollary:** AA.1 is conditionally rigorous—its rigor depends on an assumption that cannot itself be empirically verified.
2. **Tightness of  $\epsilon_\Phi$  is completely open.** The derived upper bound contains three sources of slack: (a) one application of the triangle inequality; (b) conservativeness of Hoeffding for subgaussian distributions; (c) worst-case estimation of the Lipschitz constant  $L_\phi$ . Resolving Open Problem ?? requires establishing a matching lower bound.
3. **Circularity of the Situs embedding regularity assumption AA.1.3.** This assumption requires the Situs metric to have Lipschitz constant  $\delta_{\text{situs}}$ , but this constant itself depends on the choice of representation space  $\phi$ —which is part of the mapping  $\Phi$ . This introduces a potential circular definition risk.

**Open Problem 3.2** (Tightness of  $\epsilon_\Phi$ ). *Determine the minimax-optimal  $\epsilon_\Phi^*(n, \eta, \dim(\mathcal{X}))$ —the minimum possible approximation error when approximating alignment quality via SCX auditing, as a function of dataset size, noise level, and input dimension.*

## 3.2 RLHF Preference Noise Detection (Theorem AA.2)

**Theorem 3.3** (RLHF Audit Theorem—Condition-by-Condition Verification). *Applying the SCX audit pipeline (Theorems T1, T3, T5) to alignment*

datasets  $\mathcal{D}_{\text{align}}$ , the preference Cercis Score  $S_{\text{pref}}$  satisfies:

- (i) **T1 condition (bounded measurability)**:  $S_{\text{pref}} \in [0, \frac{1}{2}]$ , and there exists a computable statistic  $T_{\text{pref}}$  detecting  $\eta > 0$  with power  $1 - \delta$ , requiring  $n = \Omega(\frac{1}{\eta^2} \log \frac{1}{\delta})$ . [Conditionally Rigorous]
- (ii) **T3 condition (unidentifiability)**: Value pluralism and annotation noise are indistinguishable in the distribution of the preference Cercis Score. There exist two data-generating processes producing exactly the same marginal distribution of  $S_{\text{pref}}$  but with different causal origins. [Rigorous]
- (iii) **T5 condition (Lipschitz)**: The expected information gain  $\text{IG}(x)$  from querying preference labels is  $L_{\text{IG}}$ -Lipschitz in  $S_{\text{pref}}$ , with  $L_{\text{IG}} = 2 \log(\frac{1+2s_{\text{max}}}{1-2s_{\text{max}}})$  on compact subsets  $[0, s_{\text{max}}]$  ( $s_{\text{max}} < \frac{1}{2}$ ). [Conditionally Rigorous]

### 3.3 Alignment Tax (Theorem AA.3)

**Theorem 3.4** (Alignment Tax Lower Bound). *The minimum fraction of resources that must be allocated to auditing (rather than training) to achieve reliable alignment is*

$$\tau_{\text{align}} \geq \frac{\eta}{1 - \eta} \cdot \frac{\text{cost}(\text{audit})}{\text{cost}(\text{train})}.$$

### 3.4 Value–Noise Indistinguishability (Theorem AA.4)

**Theorem 3.5** (Value Uncertainty vs. Noise Indistinguishability). *We explicitly construct two preference-generating processes:*

- **World A (noise-dominated)**: True preferences are unanimous ( $\tau(x_0) = 1$ ), but observations are flipped with probability  $\eta$ .
- **World B (value-pluralism-dominated)**: No annotation noise, but the population is split with proportions  $\frac{1}{2} + \eta$  and  $\frac{1}{2} - \eta$ .

*With a refined multi-annotator construction, the joint distributions over all observable variables are identical in both worlds. This establishes the Alignment Corollary of Theorem 3 with full measure-theoretic rigor. [Rigorous]*

## 4 Conclusion

AA-Theorem establishes that AI alignment auditing reduces to SCX data quality auditing through an explicit mapping  $\Phi$  with an  $\epsilon_\Phi$  approximation bound. RLHF preference noise satisfies all SCX regularity conditions, enabling the full audit pipeline. The alignment tax quantifies the resource cost of reliable alignment, and the value-noise indistinguishability result proves that genuine value pluralism is statistically indistinguishable from annotation noise — a fundamental limit on alignment verification.

## References

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